

BIMM 143 Homework: Protein Drug Interaction Function

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Overview

In this assignment, I created a function to analyze protein structures from PDB IDs. The function reads PDB files, extracts B-factors from chain A alpha carbons, and plots them. This generalizes the original repetitive code into a reusable function.

Setup

```
library(bio3d)
```

Function

```
# Function: analyze_protein_set
#
# Inputs:
#   pdb_ids = a vector of PDB IDs (e.g., c("4AKE", "1AKE", "1E4Y"))
#   chain   = chain to analyze (default "A")
#   elety   = atom type to keep (default "CA")
#
# What the function does:
#   Reads each PDB structure, trims to the selected chain and atom type,
#   extracts B-factors, and plots the B-factor profile for each protein.
#
# Output:
#   Returns a list of B-factor vectors (one per protein) and produces plots.

analyze_protein_set <- function(pdb_ids, chain = "A", elety = "CA") {

  results <- list()

  for (id in pdb_ids) {

    pdb <- read.pdb(id)
    pdb_trim <- trim.pdb(pdb, chain = chain, elety = elety)
    b_factors <- pdb_trim$atom$b

    plotb3(
      b_factors,
      sse = pdb_trim,
      typ = "l",
      ylab = "B-factor",
      xlab = "Residue Position",
      main = paste("B-factor Profile for", id)
    )
  }
}
```

```

    )

    results[[id]] <- b_factors
  }

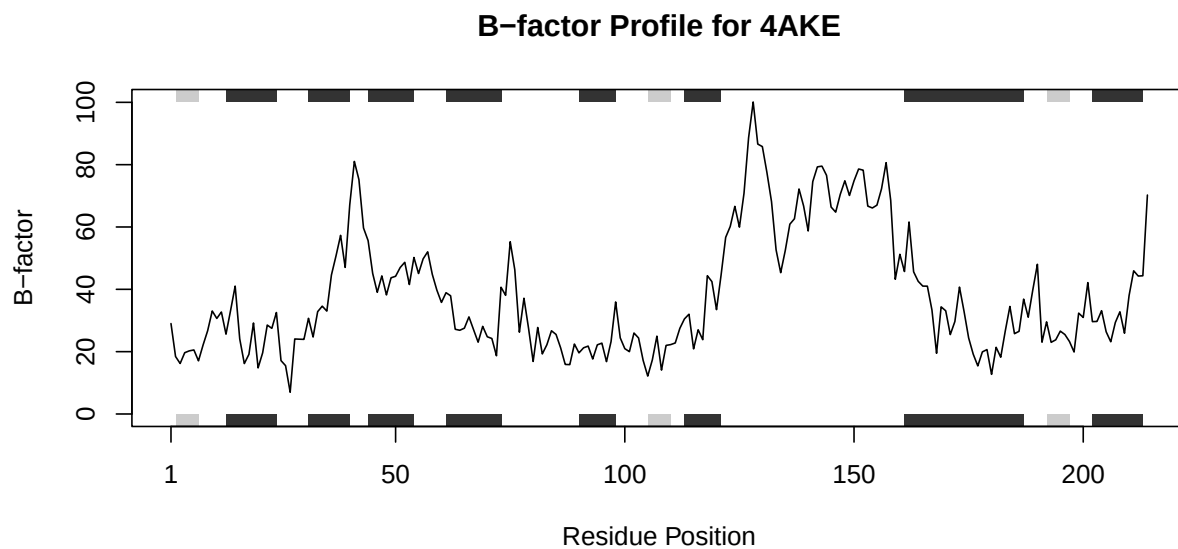
  return(results)
}

```

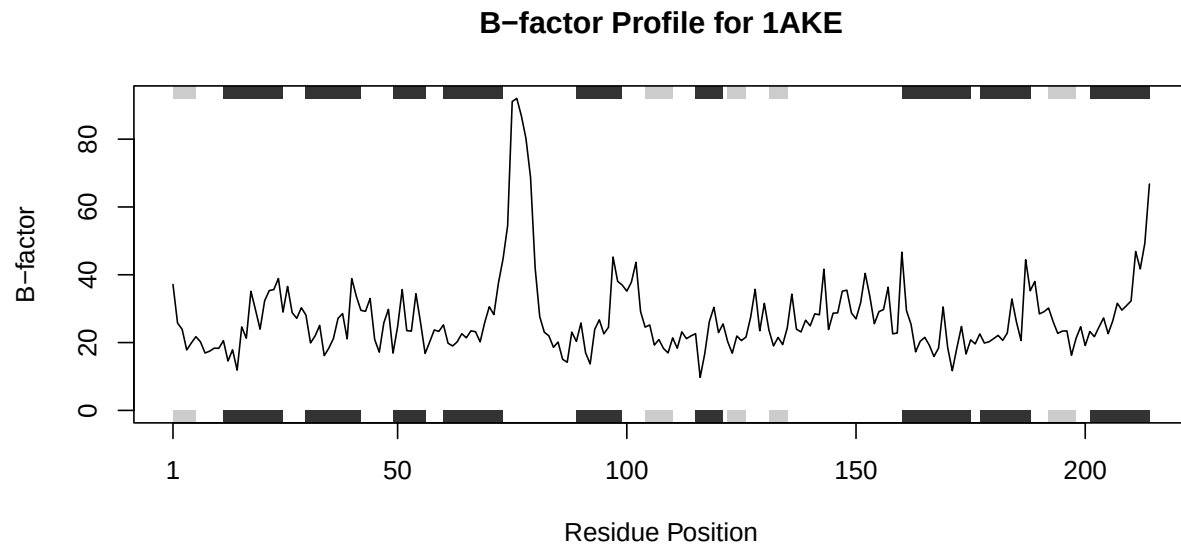
Example Usage

```
protein_results <- analyze_protein_set(c("4AKE", "1AKE", "1E4Y"))
```

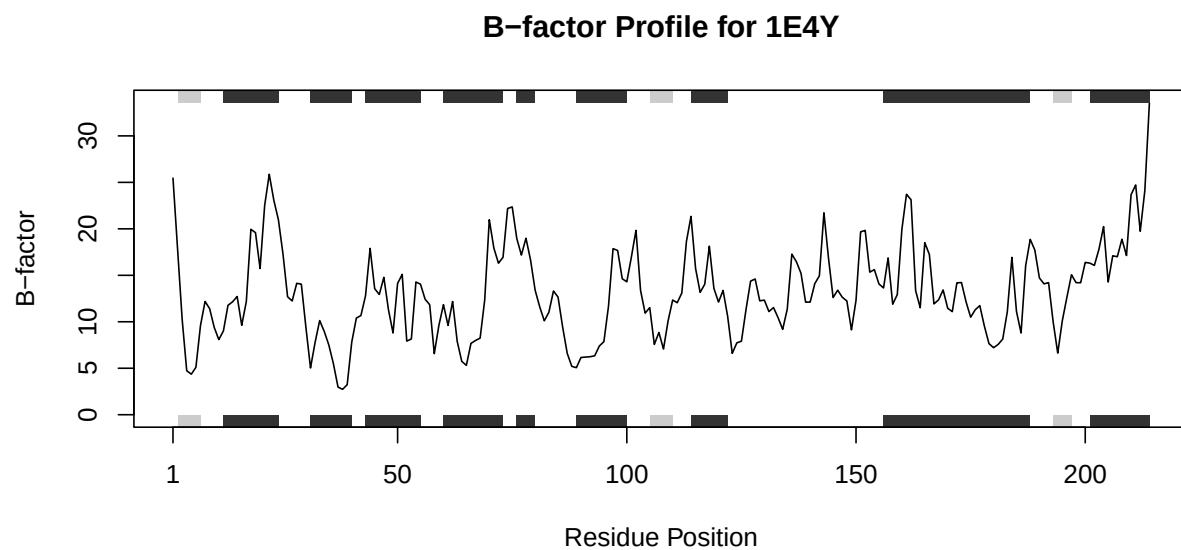
```
## Note: Accessing on-line PDB file
```



```
## Note: Accessing on-line PDB file
## PDB has ALT records, taking A only, rm.alt=TRUE
```



Note: Accessing on-line PDB file

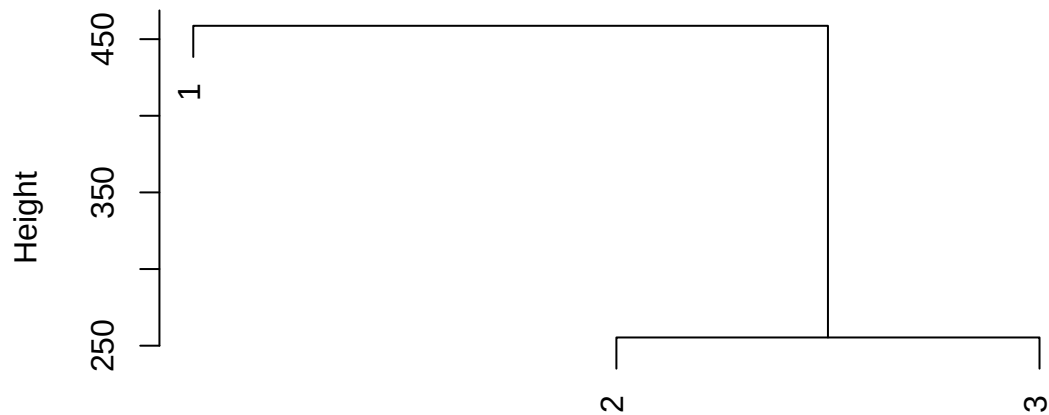


Comparison of Proteins

```
bf_mat <- rbind(
  protein_results[["4AKE"]],
  protein_results[["1AKE"]],
  protein_results[["1E4Y"]]
)

hc <- hclust(dist(bf_mat))
plot(hc, main = "Clustering of Protein B-factor Profiles")
```

Clustering of Protein B-factor Profiles



```
dist(bf_mat)
hclust (*, "complete")
```

The dendrogram above shows the similarity between proteins based on their B-factor trends. Proteins that cluster closer together have more similar flexibility patterns.

Discussion

Inputs

The function takes a vector of PDB IDs and optional arguments for chain and atom type.

What the function does

It reads each protein structure, extracts B-factors for chain A alpha carbons, and plots the B-factor profiles. This removes repetitive code and allows analysis of any number of proteins.

Output

The function produces B-factor plots and returns a list of B-factor vectors that can be used for further analysis such as clustering.